

ANSHUL DHIMAN

AI / ML Engineering Student | B.Tech CSE (AI & ML) @ LPU

Hamirpur, HP, India | **Email:** anshul.dhiman.ml@gmail.com | **GitHub:** github.com/anshuldhiman-ai | **LinkedIn:** linkedin.com/in/anshul-dhiman-ai

OBJECTIVE & FOCUS

2nd-year B.Tech CSE (AI & ML) student focused on building practical, hardware-accelerated computer vision applications, local-first LLM architectures, and robust machine learning pipelines. Experienced with PyTorch, YOLOv8, OpenCV, ONNX Runtime, FastAPI, and Docker.

TECHNICAL SKILLS

AI & Computer Vision: PyTorch, YOLOv8, OpenCV, ONNX Runtime, scikit-learn, RAG, Hugging Face
Languages & Backend: Python, C, C++, SQL, FastAPI, Flask, REST APIs, Docker, WebSockets
Data & Tools: Pandas, NumPy, MongoDB, SQLite, Git, GitHub, Linux, CI/CD pipelines

FEATURED PROJECTS

NeuralRAG – Evaluated Multi-Agent RAG Pipeline | *Python, Qdrant, BGE Reranker, Llama 3.2, Ragas*

- Built an automated retrieval-augmented generation pipeline with dense vector retrieval and BGE cross-encoder re-ranking.
- Evaluated with Ragas benchmark framework achieving 0.91 Faithfulness, 0.89 Answer Relevance, and 0.93 Context Recall.

Currency Counter using OpenCV & YOLOv8 | *Python, OpenCV, YOLOv8, ONNX Runtime, PyTorch*

- Trained a custom YOLOv8 model on 1,450 annotated currency images detecting all 7 Indian currency note denominations.
- Achieved 98.4% mAP@0.5 and 45+ FPS inference speed using hardware-aware ONNX Runtime execution (GPU/NPU fallback).
- Implemented an IoU temporal tracking algorithm eliminating object flickering and preventing duplicate counting on webcam feeds.

Batua – Personal AI Finance Manager | *Python, FastAPI, MongoDB, SQLite, Ollama, scikit-learn*

- Engineered a local-first privacy finance manager with dual-database auto-failover (MongoDB primary to SQLite fallback).
- Integrated an offline Llama 3.2 assistant via Ollama and a custom NLP parser for natural language expense tracking with zero cloud latency.

FormatFlow – Containerized Document Engine | *Python, Flask, PyMuPDF, Docker, python-docx, LibreOffice*

- Developed a 4-stage fallback conversion engine supporting 14+ input and 15+ output formats inside Docker container.

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML)

Lovely Professional University, Phagwara, Punjab

2025 – 2029

Current 2nd Year

Intermediate (Class XII) & Matriculation (Class X)

Montessori Cambridge School, Pathankot

2023 – 2025

CERTIFICATIONS & ACHIEVEMENTS

- **Infosys Springboard:** Programming Fundamentals using Python (Part 1 & 2)
- **Infosys Springboard:** Introduction to Artificial Intelligence & Neural Networks
- **Saylor Academy:** CS105 Introduction to Python Programming